Universidade do Minho

Ano Letivo: 2025/26

Databases



Data, Information, and Knowledge
Databases
The Database Development Lifecycle

Overview

In this unit we will:

• Begin the study of data systems, understanding how they emerged and their importance in the current context.

• Analyze their practical application and relevance in the real world, both at an organizational and technological level.

• Explore the fundamental concepts that any student or professional in the field of databases should master.

• Understand and clearly distinguish the concepts of data, information, and knowledge.

• Study, at a later stage, Database Management Systems (DBMS).

• Pay special attention to the storage structures used by these systems, namely the databases themselves.

Summary

1 Introduction

2 Data, Information, and Knowledge

4 Databases and Database Management Systems

5 The Database Development Life Cycle

"Information has gone from scarce to superabundant. That brings huge new benefits, but also big headaches"

Kenneth Cukier, in "Data, data everywhere – a special report on managing information", The Economist, February 27th, 2010

Introduction

- The amount of available information is growing exponentially.
- Data and information play a fundamental role in organizations.
  – Example: support for decision-making.
- It is essential to use tools that simplify data management and enable the extraction of useful information.

Introduction

Management in the recent past...







Introdução

Currently...



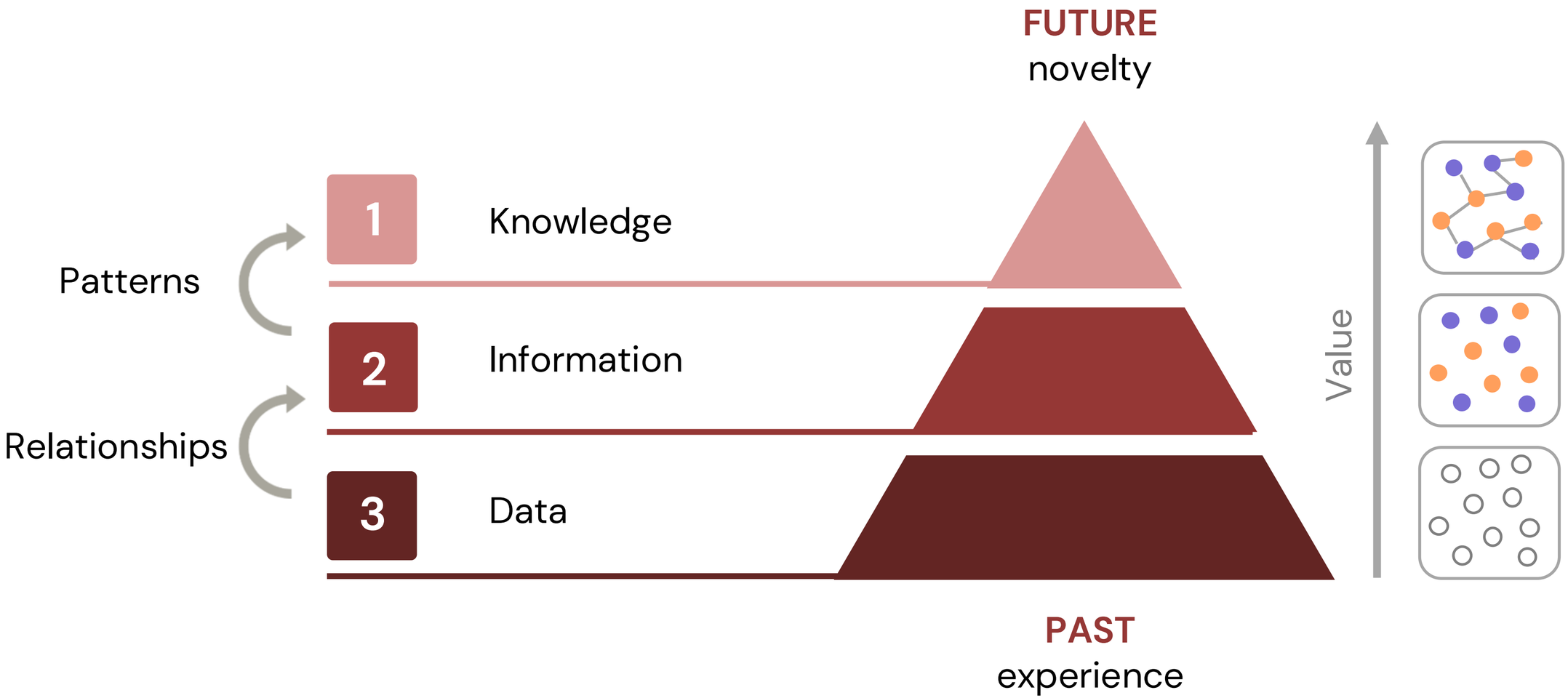


Problem: Data storage and management

Bigger problem: Turning data into information

It requires correct modeling of information!

The three essential pillars of any analysis and decision-making process:



Data, information or knowledge?

Dados (*Data*) – A set of discrete facts about a given set of events that do not present any particular structure or organization; they are the most basic elements of information — its 'atoms'.

- *There are 32 students enrolled in the course.*
- *The customer made 5 purchases this month.*

Data, information or knowledge?

Informação (Information) – it emerges when the context of data is defined and typically answers questions such as ‘What?’, ‘Why?’, ‘Where?’, ‘When?’ or ‘Who?’; it can be seen as a package of structured data that adds meaning by providing context and significance.

– *Umbrella sales increased by 40% today due to heavy rain.*
– *The customer is considered frequent, as he made 5 purchases this month.*

Data, information or knowledge?

Conhecimento (Knowledge) – the application of acquired experience and observations about the available information makes it possible to answer the question ‘How?’.

– When people leave the house with sun, they rarely bring an umbrella with them.

Databases

• Databases are fundamental components of computer systems and are present in most software applications.

• Regardless of the type of application, they play an essential role in system operation.

• Their main function is to ensure the structured and efficient storage of information.

• Without databases, a large portion of modern applications would not be able to operate consistently and reliably.

Databases

"In general, any collection of data is a Database (DB). Nowadays, the term DB is mainly used to refer to computer-based databases.

A database is a shared collection of logically structured/interrelated data, developed to meet an organization's information needs.

USE EXAMPLES:

- Supermarkets/Hypermarkets;
- Bookstores;
- E-commerce sites;
- Banks;
- Gyms;
- Streaming platforms (Netflix/HBO);
- University;
- Etc.

Databases

• A construção de uma base de dados baseia-se no desenvolvimento de um **modelo de dados**.

• O modelo de dados representa a forma como a informação é estruturada e organizada.

• É normalmente expresso através de **diagramas (esquemas)** que traduzem os requisitos da aplicação a diferentes níveis de abstração.

• Para ser eficaz, o modelo deve refletir corretamente a realidade e as necessidades de armazenamento da aplicação.

• A criação de uma base de dados deve seguir uma abordagem **metódica e estruturada**, utilizando padrões e boas práticas de modelação.

Database Management Systems

Databases are manipulated using a **Database Management System (DBMS)** or **Database Management System (DBMS).**

A DBMS is a software system that manages and controls all operations performed on database content, such as access control, persistence, and data manipulation, among others. It ensures interaction between applications and databases.

Whenever a user creates a valid SQL (Structured Query Language) statement, the DBMS is responsible for its interpretation, execution, and the corresponding response.

Database Management Systems

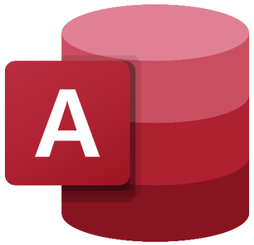










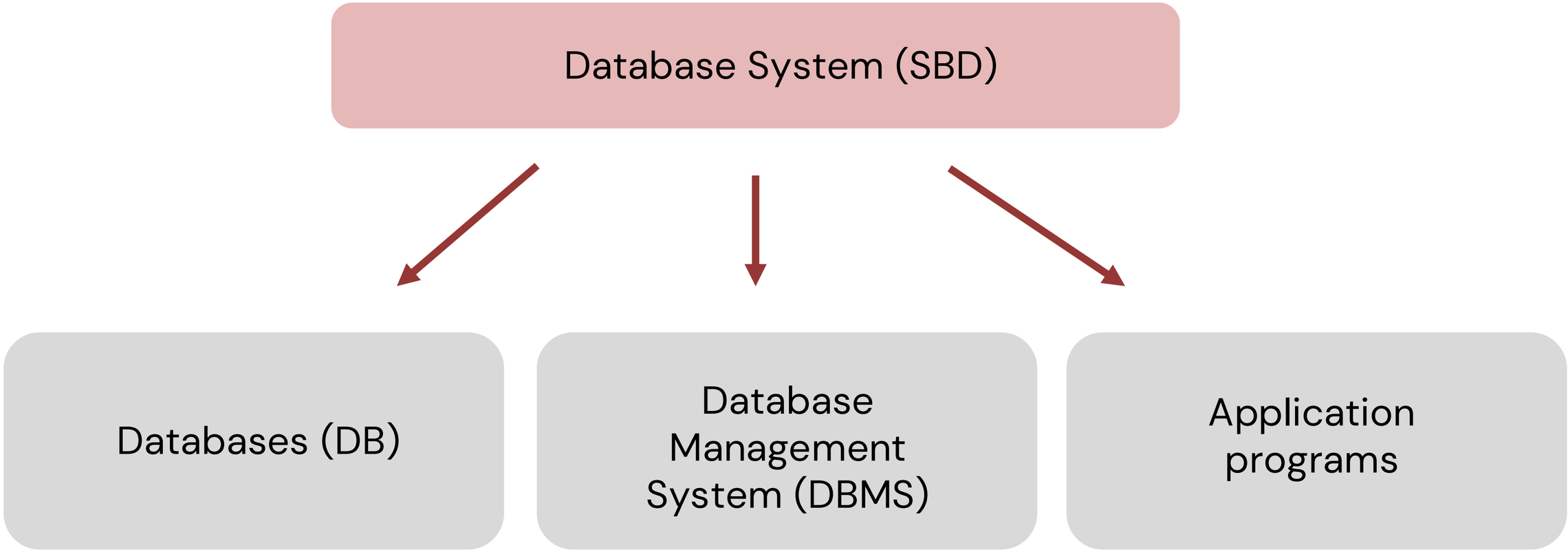




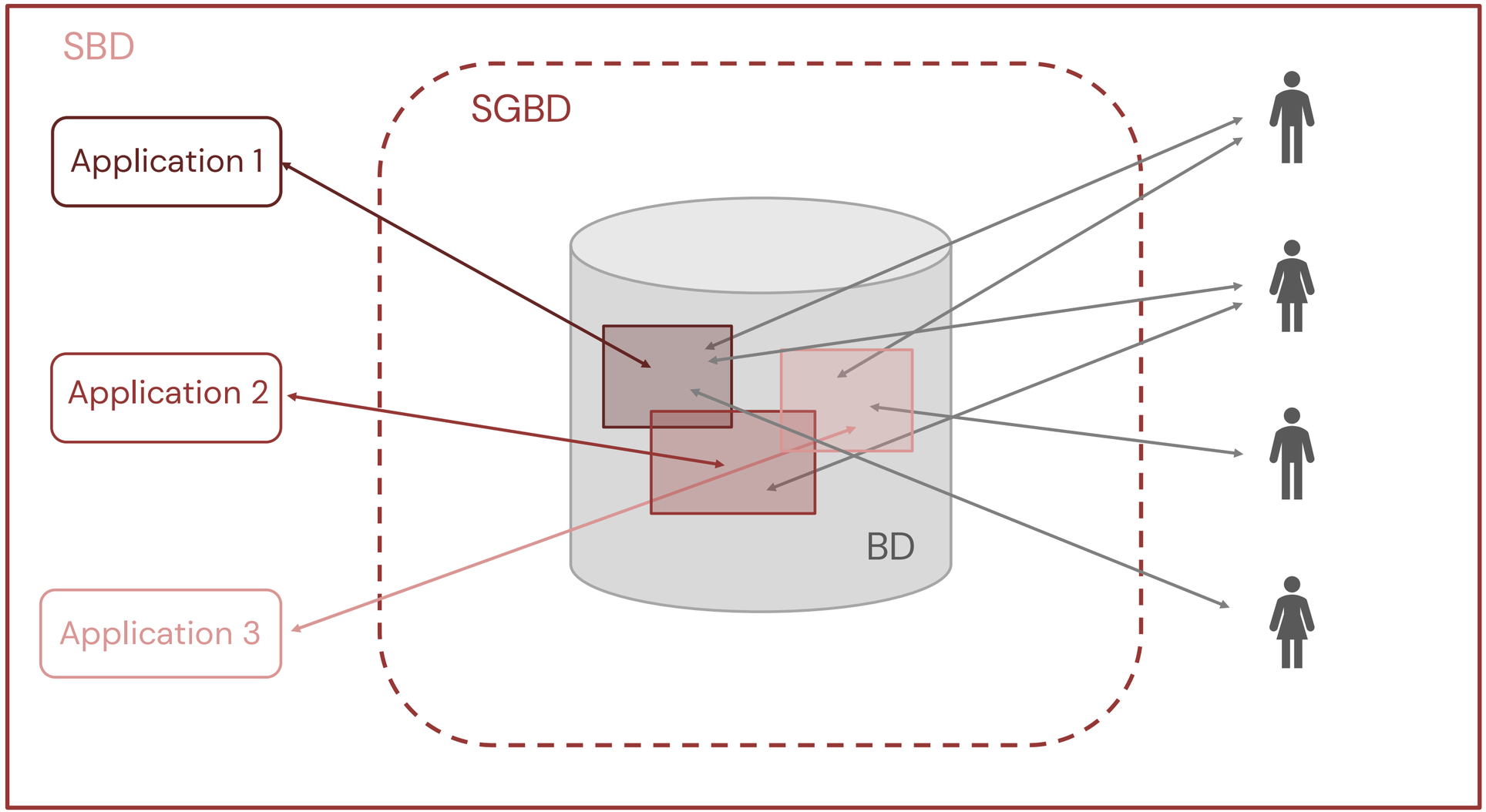




Database Systems



Database Systems



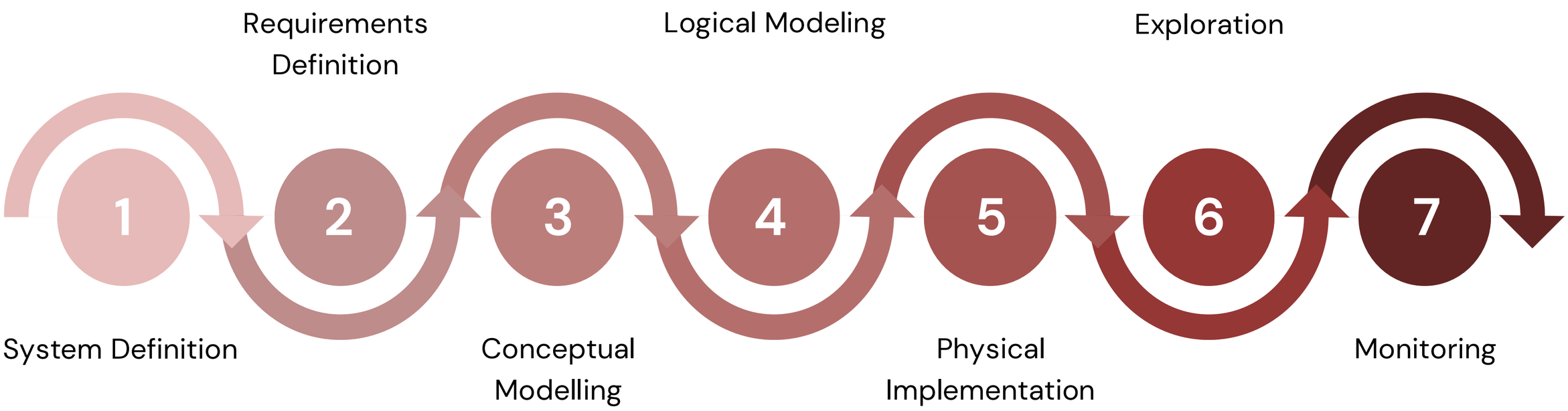
Database Systems

Why should we use an SBD?

- Greater control of data redundancy;
- Data sharing;
- Restriction of unauthorized access;
- Improved data integrity and security;
- Data persistence;
- Enforcement of data integrity constraints;
- Enables data recovery and backup;
- Improved decision-making processes;
- (...)

Database Development Lifecycle

A practical blueprint/guide that ensures the orderly and proper execution of database development tasks, reducing the likelihood of errors.



Database Development Lifecycle: System Definition

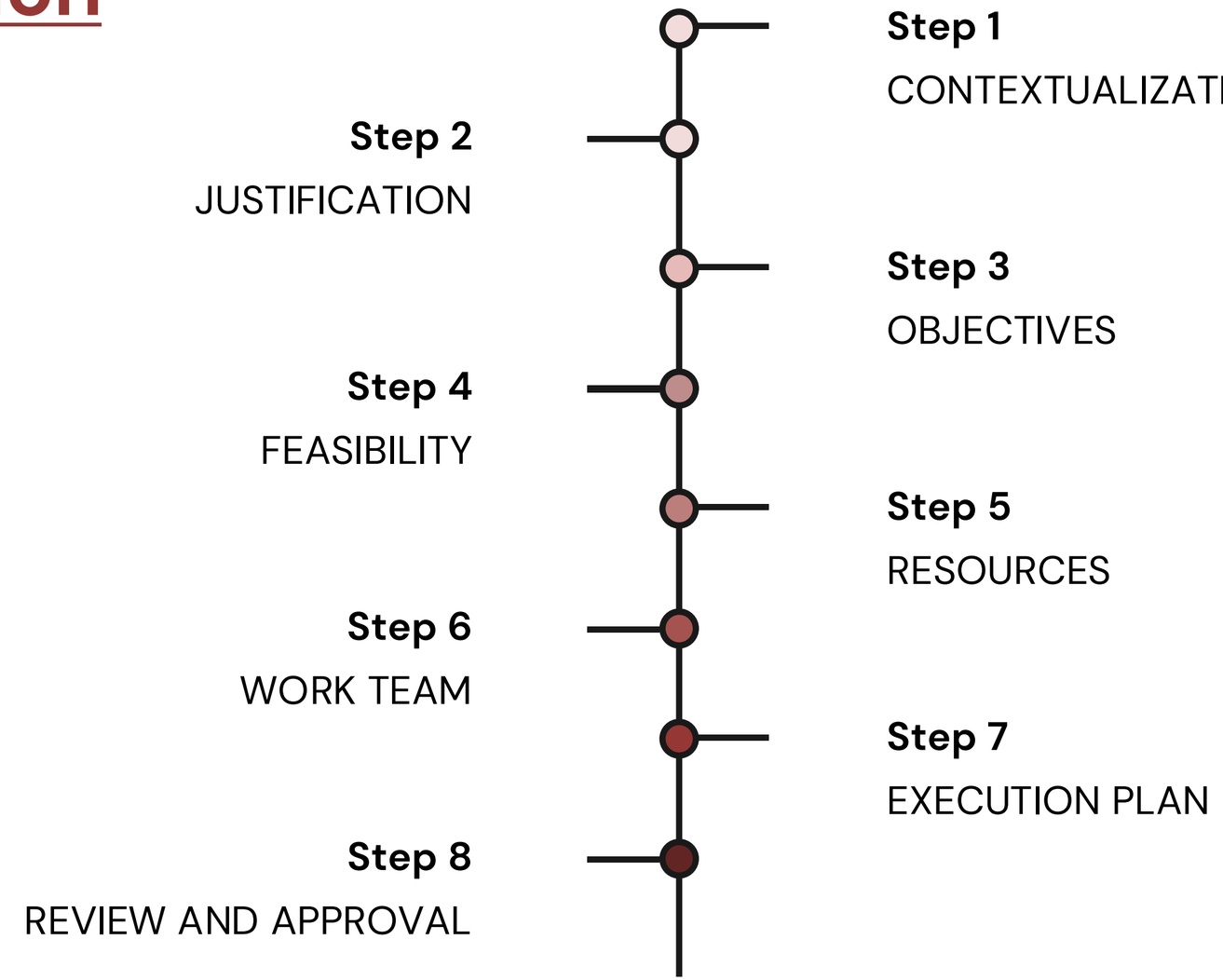
Before starting requirements gathering or database modeling, it is essential to clearly understand why the database is being built.

• Defining the context and objectives is essential to guide the development process.

• Conducting a feasibility analysis helps assess the need for and usefulness of the solution.

• A good understanding of the problem improves:
– Requirements gathering
– Team communication

Database Development Lifecycle: System Definition



PHASE 1: System Definition

Contextualization

• Contextualization should precede database development.

• It is essential to understand:
  – The organization
  – The operational environment
  – The processes involved
  – The stakeholders

• The boundaries of the problem should be clearly defined, avoiding the inclusion of areas or processes that are not relevant.

• Good contextualization helps focus development on what is truly necessary.

PHASE 1: System Definition

Contextualization

Case Study:

To facilitate understanding of the database system development process, the methodology will be described using a realistic example: the company “TechSol” and the “Desporto Total Training Center” (CTDT), covering each of its working phases in detail.
The company “TechSol” and the “Desporto Total Training Center” are purely fictitious and were created solely for the development of a practical case that can serve as a working basis for database system development.
This case will be developed and explored gradually, in line with the development of the underlying database system. Throughout the classes, the study elements required to support the specific case study requirements and the various stages of the database system development lifecycle will be introduced.

PHASE 1: System Definition

Contextualization

Questão 1 : Contextualize the company “TechSol” and the “Desporto Total Training Center”. At this stage, it is important to define where the company/institution is located, when it was founded, what purpose it serves, who the people involved are, etc..

“TechSol” is a technology company headquartered in Lisbon, specializing in the development of customized data management and information systems solutions since 2005. Recently, the company’s CEO, Engineer Teresa Maia, received a proposal from the “Desporto Total Training Center” (CTDT), located in the city of Setúbal, to create a data management system that would optimize the center’s operations.

The CTDT is a renowned institution in the sports sector, specifically in personalized training, and was founded in 1990 by former Olympic athlete Carlos Silva. The center’s board is composed of President Carlos Silva, Technical Director André Santos, and Training Coordinator Maria Oliveira.

PHASE 1: System Definition

Justification

- The rationale for the database development process should outline, in general terms, the reasons for the need for the database, briefly presenting the systems and processes it will support, how they will interact with it, and what information the database will store.

- Additionally, it should also clarify how the database will integrate into the organization's computing infrastructure.

PHASE 1: System Definition

Justification

Questão 2 : Justify why the company “TechSol” should develop a DBMS for the “Desporto Total Training Center”.

Currently, CTDT faces several challenges such as the increasing demand for sports training services, limitations in physical and human resources, and difficulties in managing schedules and athletes’ data. Given these challenges, CTDT’s management decided to seek a technological solution to improve its operations. “TechSol” was contracted to develop an information system that would cover everything from scheduling training sessions to monitoring athletes’ health. The proposed system should include functionalities such as recording athletes’ medical information, registering consultations with specialists from different areas of sports health, and tracking training routines.

PHASE 1: System Definition

Objectives

• Objectives define what is intended to be achieved with the development of the database.

• They represent the expected outcomes of the system to be built.

• They serve as guidance for all phases of the project.

• They should be aligned with the organization’s needs.

• They make it possible to evaluate, at the end, whether the developed solution fulfills its intended purpose.

PHASE 1: System Definition

Objectives

Questão 3 : List some of the objectives defined by the management team to meet the needs of the "Desporto Total Training Center".

Taking into account their experience and knowledge in the field, CTDT's management defined a set of objectives they aim to achieve:

1. Organize the center's business model;
2. Improve its capacity to manage and record athletes, coaches, training sessions, consultations, and specialists;
3. Efficiently manage the scheduling of training sessions and athletes' examinations;

- (...);

PHASE 1: System Definition

Feasibility

Analyzing the feasibility of building a database should be supported by a brief study carried out in a preliminary phase of requirements gathering (or earlier), whose main objective is to provide answers to questions such as:

– What are the main objectives to be achieved with the development of the database?
– Can the database be built according to the established schedule and with the available resources?
– Are the systems whose operation will be supported by the database adequately prepared?
– (...)

PHASE 1: System Definition

Feasibility

Questão 4 : Analyze the feasibility of the previously defined database system, stating the potential key effects of its possible adoption.

The “Desporto Total Training Center” believes that by having an effective data management and information systems solution, it will be able to:

- Schedule training sessions more efficiently, ensuring that resources are used appropriately and that there is a variety of training options for athletes;
- Ensure the recording of all clinical information associated with an athlete in order to improve the therapeutic decision-making process;
- Increase the number of registered athletes by approximately 15% compared to the previous year;
- (...);

PHASE 1: System Definition

Resources

When developing a database, it is essential to identify the resources needed to carry out the project.

Human Resources
- People involved in the planning, development and maintenance of the system.
- Elements with technical skills and knowledge of the context of the organization.
- Participation of users to ensure that the solution meets real needs.

Material Resources
- Technological means necessary for implementation.
- Support equipment and infrastructure.
- Tools that allow you to develop, run and maintain the database.

PHASE 1: System Definition

Resources

Questão 5 : List the features (function + quantity) required for the development and implementation of the defined SBD.

HUMAN

– Equipa de desenvolvimento da "TechSol";
– Diretoria do CTDT;
– Profissionais de Saúde;
– Atletas.

MATERIALS

– Hardware: 1 server + 4 machines (2 tests and 2 production).

– Software (DBMS and IS)

PHASE 1: System Definition

Work Team

- It must identify the necessary actors for the project.
- Defines responsibilities and roles of each element.
- It promotes a clear distribution of tasks.
- It facilitates communication and coordination among members.

PHASE 1: System Definition

➡ Work Team

Questão 6 : Define the work team (internal staff, external staff, others) taking into account the associated distribution of tasks.

INTERNAL PERSONNEL	EXTERNAL PERSONNEL	OTHERS
	Development team of the company "TechSol"	Athletes
- Operation of the center; - Training; - Conducting consultations; - (...);	- Requirements gathering; - System modeling; - Implementation of the system; - (...)	- Opinion surveys and evaluation of the quality of the services provided;

PHASE 1: System Definition

Execution Plan

The execution plan defines the temporal and operational organization of the database development project.

- Establishes the phases and activities to be carried out.
- Sets deadlines and task sequence.
- Identifies dependencies between activities.
- Assigns responsibilities to the team.
- Allows you to track the progress of the project.

PHASE 1: System Definition

Execution Plan

Questão 7 : How can tasks be defined and planned over time in order to meet the established deadlines?

To define and plan how the database system development process would be carried out, the board of directors of the “Desporto Total Training Center” and the “TechSol” team selected for its development established a concrete work plan, as well as its execution schedule.
Using a Gantt chart, a work plan was developed that included all stages of the database system lifecycle, defining their execution periods and the various stakeholders involved in carrying out the tasks.

PHASE 1: System Definition

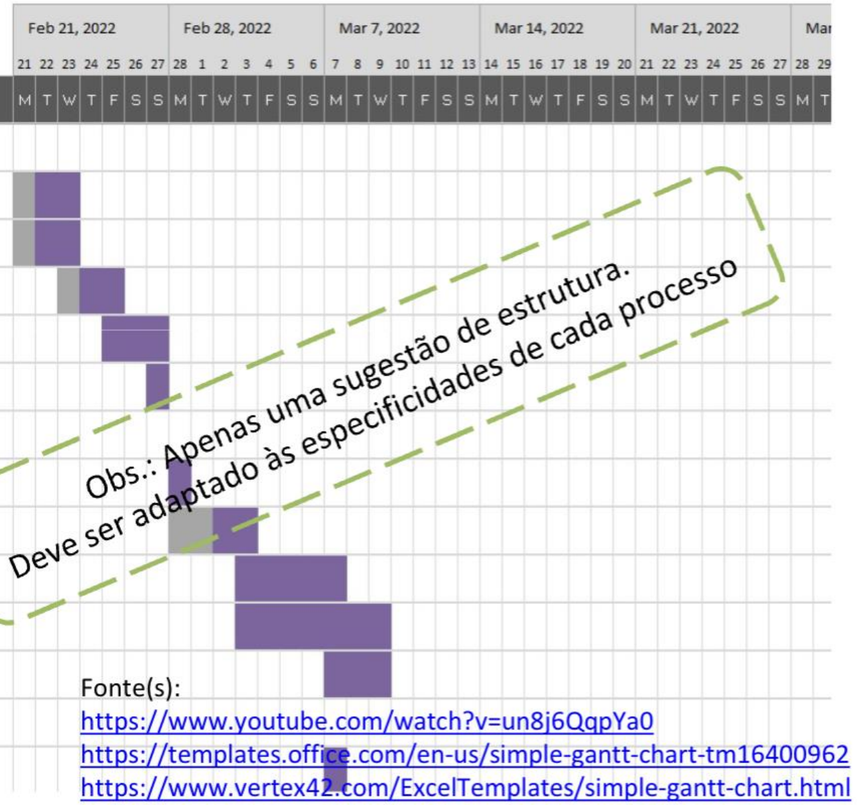
Execution Plan

Implementação do Sistema de Bases de Dados

A Mercearia da D. Acácia
D. Acácia

Início do Projeto: Mon, 2/21/2022
Semana: 1

Tarefa	Atribuída a	Progresso	Início	Fim
Definição do Sistema				
Contextualização e Fundamentação	D.Acácia	50%	2/21/22	2/23/22
Objetivos e Viabilidade	D.Acácia, Sr.Pascoal	60%	2/21/22	2/23/22
Recursos e Equipa	D.Acácia, Sr.Pascoal, Empresa	50%	2/23/22	2/25/22
Plano de Execução	D.Acácia, Sr.Pascoal, Empresa	25%	2/25/22	2/27/22
Revisão e Aprovação	D.Acácia, Sr.Pascoal, Empresa		2/27/22	2/27/22
Definição de Requisitos				
Definição do Método	Arquiteto, Engenheiro, Analista	50%	2/28/22	2/28/22
Levantamento de Requisitos	Analista	50%	2/28/22	3/3/22
Análise de Requisitos	Analista		3/3/22	3/7/22
Organização de Requisitos	Arquiteto, Engenheiro, Analista		3/3/22	3/9/22
Validação	D. Acácia, Arquiteto, Engenheiro, Analista		3/7/22	3/9/22
Modelação Concetual				
Definição do Processo de Modelação			3/7/22	3/7/22



Exemplo

Obs.: Apenas uma sugestão de estrutura.
Deve ser adaptado às especificidades de cada processo

Fonte(s):
https://www.youtube.com/watch?v=un8j6QqpYa0
https://templates.office.com/en-us/simple-gantt-chart-tm16400962
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

PHASE 1: System Definition

Review and Approval

The review and approval phase consists of a detailed verification of the work carried out throughout the project.

- It allows the identification of errors, inconsistencies, or system flaws.
- It verifies whether the requirements and objectives initially defined have been met.
- It involves analysis by the development team.
- It includes validation by users and other stakeholders.
- It may lead to the need for adjustments or improvements before final approval.

PHASE 1: Definition of the system

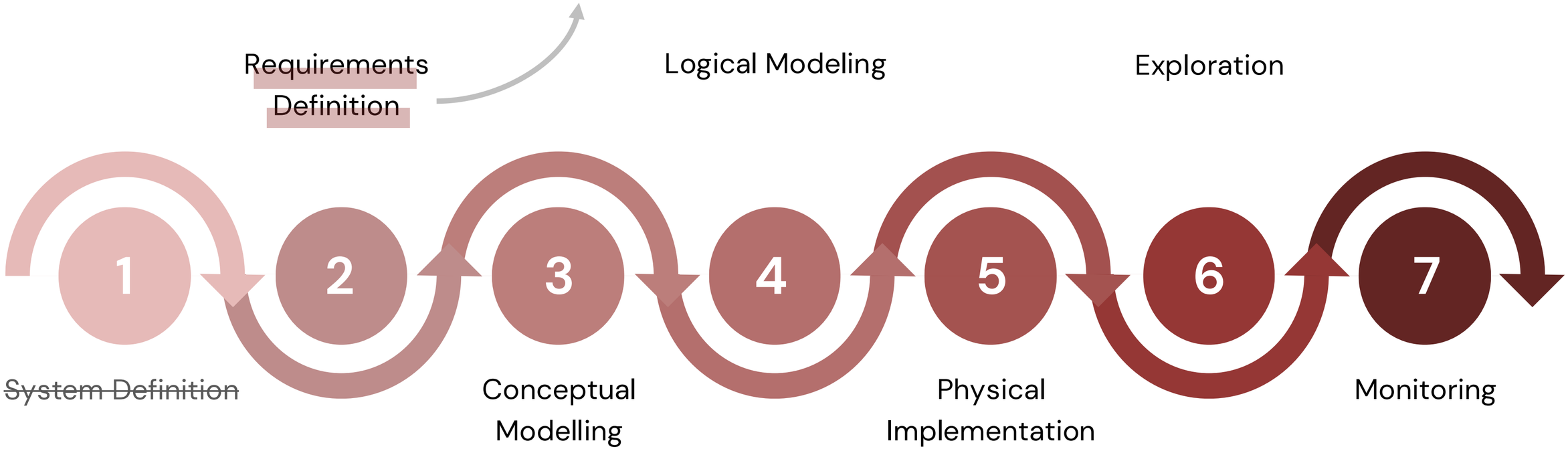
Review and Approval

Question 8 : The SBD defined was approved by the board of directors of the "Total Sports Training Center". Are there aspects to be improved or that have not been taken into consideration?

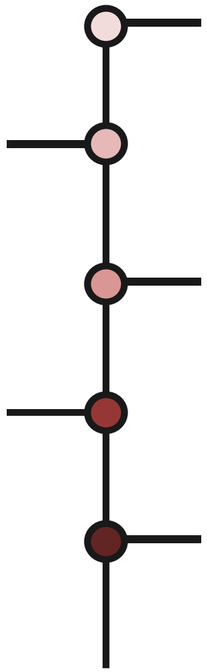
After the “TechSol” team completed the execution plan, Engineer Teresa Maia decided to contact the board of directors of the “Desporto Total Training Center” to schedule a business meeting. During this meeting, they discussed the system definition, its rationale, its main objectives, and its feasibility.
Together, the board of directors, the school director, and the CEO of “TechSol” confirmed everything that had been defined and validated the work plan developed through the Gantt chart.
The school’s administrative board decided to hire “TechSol” to proceed with the development and implementation of the proposed database system. A requirements definition meeting was then scheduled for March 4, 2024.

Lifecycle of an SBD

Gathering, analyzing and organizing requirements about the part of the organization that will be supported by SBD. The requirements must be able to describe what data is intended to be stored and possible relationships between them.



Lifecycle of an SBD: Requirements Definition



Step 1
DEFINITION OF THE METHOD

Step 2
REQUIREMENTS GATHERING

Step 3
REQUIREMENTS ANALYSIS

Step 4
REQUIREMENTS ORGANIZATION

Step 5
VALIDATION

PHASE 2: Requirements Definition

Method Definition

The “Requirements Definition” process begins with the selection of elicitation methods to support the gathering and organization of requirements. There are many techniques used to collect this information, known as fact-finding techniques:



Documentation Analysis



Interviews/Meetings



Questionnaires /Surveys



Research (state-of-the-art survey)



Observation of the organization in operation in its day-to-day life

PHASE 2: Requirements Definition

Method Definition

Question 1: According to the above withdrawal methods, identify the one (s) that should be used.

PHASE 2: Requirements Definition

Method Definition

A database developer typically uses several fact-finding techniques during a database project.

- **Interviews:** Meetings between the center's board of directors and the "TechSol" development team to understand operational processes/workflows in the different working contexts of the training center.
- **Observation of the day-to-day activities of the "Desporto Total Training Center":** Viewing/interacting with processes related to athlete services, training sessions, etc.
- **Questionnaires:** Distribution of opinion and satisfaction surveys to athletes.
- **Document Analysis:** Information related to staff contracts, equipment, events, etc.
- **Research:** Searching for relevant information in the context of the case at hand, namely from other training centers to identify possible solutions/approaches.

Requirements Document

The information obtained through fact finding is then analyzed to identify the requirements to be included in the new SBD. These requirements are described in documents called requirements specifications or requirements documents.

Nº	Date and Time	Description	Area	Source	Analyst

PHASE 2: Requirements Definition

Requirements Gathering

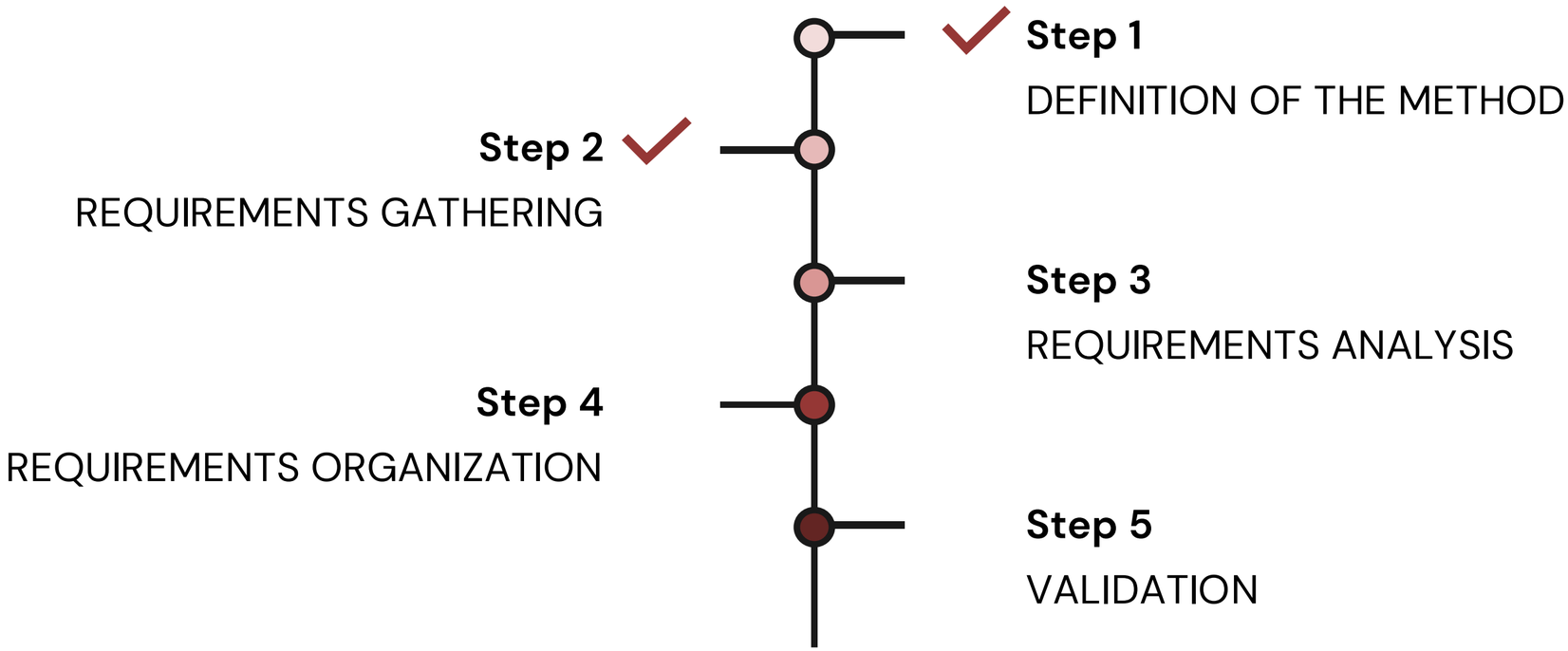
Question 3 : Identify all the requirements for the development of the database system for the defined case study — “TechSol” and the “Desporto Total Training Center”.

PHASE 2: Requirements Definition

Requirements Gathering

Nº	Date/Time	Description	Area/View	Source	Analyst
1	13/02/2024 17:00	All athletes at the center must be registered with a unique identification number.	Athletes	Centro de Treino	C.N.
2	13/02/2024 17:00	Personal data of the athletes must be stored, such as full name, date of birth, address, telephone, e-mail, and NIF.	Athletes	Centro de Treino	C.N.
3	13/02/2024 17:00	In terms of the type of profession, it is only intended to consider athletes, health professionals and coaches.	Employees	Centro de Treino	C.N.
4	13/02/2024 17:00	Each specialty must be characterized by an identification code and a detailed description.	Appointment	Centro de Treino	C.N.
5	13/02/2024 17:00	At any given moment, it should be possible to obtain the list of health professionals of a particular specialty.	Healthcare Professionals	Centro de Treino	C.N.
6	13/02/2024 17:00	Only one appointment can be held at a time in a given office and time, avoiding overlaps.	Appointment	Centro de Treino	C.N.
...	...	...	...	...	...

Lifecycle of an SBD: Requirements Definition



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